Table A1. Random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), single-motorcycle crashes, Thailand, 2016–2023 (two-year periods), partially constrained temporal structure.

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Constant [serious injury / fatal]				
Constant [2016–2019]	0.489	4.56		
Constant [2020–2023]	–0.453	–2.95		
Variables producing the same parameter value across all periods [2016–2023]				
Crash type				
Ran off road to the left and hit an object (1 if yes, 0 otherwise) [2016–2023]	0.406	2.95	–0.0655	0.0655
Vehicle mounted a traffic island (1 if yes, 0 otherwise) [2016–2023]	0.728	1.69	–0.0667	0.0667
(Standard deviation of parameter distribution; 60.9% above zero)	2.627	2.56		
Ran off road to the right (1 if yes, 0 otherwise) [2016–2023]	–0.353	–2.76	0.0578	–0.0578
Roadway characteristics				
Depressed median (1 if yes, 0 otherwise) [2016–2023]	0.436	4.64	–0.0705	0.0705
Environmental characteristics				
Dark, with street lighting (1 if yes, 0 otherwise) [2016–2023]	0.977	9.32	–0.1547	0.1547
Temporal characteristics				
Songkran festival period (1 if yes, 0 otherwise) [2016–2023]	–0.547	–3.81	0.1051	–0.1051
(Standard deviation of parameter distribution; 41.2% above zero)	2.470	4.43		
Evening (1 if yes, 0 otherwise) [2016–2023]	–0.770	–8.07	0.1232	–0.1232
Variables with temporally varying parameters				
Crash type				
Rear-end collision in the same lane (1 if yes, 0 otherwise) [2020–2023]	0.495	1.97	–0.0824	0.0824
Sideswipe (parallel movement) (1 if yes, 0 otherwise) [2020–2021]	0.858	2.22	–0.1435	0.1435
Ran off road to the left (1 if yes, 0 otherwise) [2016–2017]	–0.058	–0.25	0.0175	–0.0175
(Standard deviation of parameter distribution; 49.2% above zero)	2.795	2.09		
Ran off road to the left (1 if yes, 0 otherwise) [2018–2019]	–0.760	–4.94	0.1132	–0.1132
Ran off road to the left (1 if yes, 0 otherwise) [2020–2023]	–0.383	–3.40	0.0647	–0.0647
Loss of vehicle control (1 if yes, 0 otherwise) [2018–2023]	–1.333	–11.01	0.2185	–0.2185
Ran off road on a right bend (1 if yes, 0 otherwise) [2016–2017]	0.454	1.91	–0.0762	0.0762
Crash cause				
Speeding (1 if yes, 0 otherwise) [2016–2017]	–0.474	–3.62	0.0792	–0.0792
Speeding (1 if yes, 0 otherwise) [2020–2023]	0.679	4.71	–0.1149	0.1149
Cutting in too closely (1 if yes, 0 otherwise) [2016–2017]	–0.691	–3.03	0.1169	–0.1169
Driver fell asleep (1 if yes, 0 otherwise) [2022–2023]	1.964	4.08	–0.2796	0.2796
Drunk driving (1 if yes, 0 otherwise) [2016–2019]	–0.735	–4.16	0.1170	–0.1170
Drunk driving (1 if yes, 0 otherwise) [2020–2023]	0.693	2.92	–0.1132	0.1132
Roadway characteristics				
Crash on the main carriageway (1 if yes, 0 otherwise) [2016–2017]	–0.701	–2.76	0.1197	–0.1197
Crash on the main carriageway (1 if yes, 0 otherwise) [2018–2023]	0.433	2.03	–0.0682	0.0682
Crash on a frontage road (1 if yes, 0 otherwise) [2016–2019]	–1.432	–6.08	0.2243	–0.2243
Painted median (1 if yes, 0 otherwise) [2016–2019]	0.353	2.04	–0.0550	0.0550
Painted median (1 if yes, 0 otherwise) [2020–2023]	–0.330	–1.99	0.0558	–0.0558
Raised or earth median (1 if yes, 0 otherwise) [2016–2017]	–0.503	–3.16	0.0862	–0.0862
Barrier median (1 if yes, 0 otherwise) [2016–2017]	–0.620	–2.77	0.1069	–0.1069
Barrier median (1 if yes, 0 otherwise) [2020–2023]	–0.345	–2.28	0.0585	–0.0585
Concrete pavement surface (1 if yes, 0 otherwise) [2018–2023]	–0.446	–2.69	0.0712	–0.0712
(Standard deviation of parameter distribution; 42.8% above zero)	2.450	3.83		
Environmental characteristics				
Dark, no street lighting (1 if yes, 0 otherwise) [2016–2017]	1.639	7.63	–0.2598	0.2598
Dark, no street lighting (1 if yes, 0 otherwise) [2018–2023]	1.128	8.23	–0.1733	0.1733
Temporal characteristics				
New Year festival period (1 if yes, 0 otherwise) [2016–2017]	–0.537	–4.17	0.0916	–0.0916

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
New Year festival period (1 if yes, 0 otherwise) [2018–2019]	–1.366	–4.06	0.1718	–0.1718
(Standard deviation of parameter distribution; 31.6% above zero)	2.858	2.76		
New Year festival period (1 if yes, 0 otherwise) [2020–2023]	–0.640	–4.82	0.1028	–0.1028
(Standard deviation of parameter distribution; 31.6% above zero)	1.335	2.66		
Peak hours (1 if yes, 0 otherwise) [2016–2017]	0.403	2.38	–0.0677	0.0677
Peak hours (1 if yes, 0 otherwise) [2018–2023]	–0.209	–1.88	0.0337	–0.0337
Region				
Eastern region (1 if yes, 0 otherwise) [2016–2021]	0.764	4.53	–0.1206	0.1206
Northeastern region (1 if yes, 0 otherwise) [2016–2021]	0.605	6.43	–0.0982	0.0982
Northeastern region (1 if yes, 0 otherwise) [2022–2023]	1.045	5.75	–0.1698	0.1698
Northern region (1 if yes, 0 otherwise) [2020–2023]	–0.350	–2.90	0.0596	–0.0596
Western region (1 if yes, 0 otherwise) [2020–2021]	1.254	4.18	–0.2037	0.2037
Heterogeneity in the means of random parameters				
Concrete pavement surface [2018–2023]: Eastern region [2022–2023]	–2.220	–2.90		
Songkran festival period [2016–2023]: Southern region [2016–2023]	–1.037	–3.18		
Songkran festival period [2016–2023]: Northern region [2016–2019]	–1.079	–2.75		
Model statistics				
Number of observations		7,325		
Number of estimated parameters		52		
Log-likelihood at zero, LL(0)		–5077.30		
Log-likelihood with constant only, LL(C)		–5074.03		
Log-likelihood at convergence, LL(β)		–4526.94		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$		0.108		
AIC / BIC		9,157.9 / 9,516.6		
Random parameter distribution; simulation		Normal; 2,000 Halton draws		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median, one motorcycle. Periods in brackets share one parameter value; for a variable with temporally varying parameters, periods not shown have a parameter of zero. Temporal structure from the partially constrained procedure in the fixed-parameter binary logit (Stage 1): adjacent periods merged when a likelihood ratio test does not reject equal parameters at the 90% level. The stage at which the structure was chosen (fixed-parameter or random parameters model) is the one with the lower BIC among models not rejected at the 90% level against the fully unconstrained models (Table 3; all candidates in Table D1). Random parameters (at most eight), heterogeneity in means (at most three terms) and in variances (at most one term) were searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level (a parameter is kept random only while its standard deviation is significant). A variable used in a heterogeneity function enters only the random parameter shown, for crashes in the shared periods, and is removed from the main utility. Marginal effects are average changes in the simulated unconditional probability (2,000 Halton draws) when the indicator switches from 0 to 1, averaged over crashes in the parameter's periods; for random parameters they include the heterogeneity terms. PDO/minor = –(serious/fatal). Percentage above zero evaluated at the heterogeneity variables equal to zero.